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MILITARY SPECIFICATION

MOLDING PLASTICS, ELECTRICAL, THERMOPLASTIC

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the basic properties of thermoplastic molding compounds used to mold electric and electronic component parts.

1.2 Classification. The thermoplastic molding compounds governed by this specification shall be of the following types, as specified (see 6.2):

Types	Description
PT-F	Thermoplastic polyester (polytetramethylene terephthalate or poly 1, 4 butylene terephthalate) compound, flame resistant
GPT-15F	Thermoplastic polyester (polytetramethylene terephthalate or poly 1, 4 butylene terephthalate) compound, 15 percent glass reinforced, flame resistant
GPT-20F	Thermoplastic polyester (polytetramethylene terephthalate or poly 1, 4 butylene terephthalate) compound, 20 percent glass reinforced, flame resistant
GPT-30F	Thermoplastic polyester (polytetramethylene terephthalate or poly 1, 4 butylene terephthalate) compound, 30 percent glass reinforced, flame resistant
GAT-22F	Thermoplastic, polyarylether compound, 22 percent glass reinforced, flame resistant
GET-30F	Thermoplastic polyester (polyethylene terephthalate) compound, 30 percent glass reinforced, flame resistant
GMT-40F	Thermoplastic polyester (polytetramethylene terephthalate or poly 1, 4 butylene terephthalate) compound, 40 percent glass and mineral reinforced, flame resistant
GLT-10F	Thermoplastic polyetherimide compound, 10 percent glass reinforced, flame resistant

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Electronics Systems Command (ELEX-8111), Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

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Types	Description
GLT-20F	Thermoplastic polyetherimide compound, 20 percent glass reinforced, flame resistant
GLT-30F	Thermoplastic polyetherimide compound, 30 percent glass reinforced, flame resistant
GST-40F	Thermoplastic polyphenylene sulfide compound, 40 percent glass reinforced, flame resistant

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-14	Molding Plastics and Molded Plastic Parts, Thermosetting.

STANDARD

FEDERAL

FED-STD-595	Colors.

(Copies of specifications, standards, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications, form a part of this specification to the extent specified herein.

NATIONAL TECHNICAL INFORMATION SERVICE

AD297457	Procedure for Determining Toxicity of Synthetic Compounds

(Application for copies should be addressed to Department of Commerce, National Technical Information Service, 5285 Port Royal Road, Springfield, Virginia 22151.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 149	Dielectric Breakdown Voltage and Dielectric Strength of Electrical Insulating Materials at Commercial Power Frequencies, Tests for.
D 150	A-C Loss Characteristics and Dielectric Constant (Permittivity) of Solid Electrical Insulating Materials, Tests for.
D 229	Rigid Sheet and Plate Materials Used for Electrical Insulation, Testing.
D 256	Impact Resistance of Plastics and Electrical Insulating Materials, Tests for.
D 495	High-Voltage, Low-Current, Arc Resistance of Solid Electrical Insulating Materials, Test for.

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D 570	Water Absorption of Plastics, Test for.
D 638	Tensile Properties of Plastics, Test for.
D 648	Deflection Temperature of Plastics Under Flexural Load, Test for.
D 695	Compressive Properties of Rigid Plastics, Test for.
D 790	Flexural Properties of Plastics, Test for.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

UNDERWRITERS LABORATORY INCORPORATED

UL Standard 94	Tests for Flammability of Plastic Materials for Parts in Devices and Appliances.

(Application for copies should be addressed to the Underwriters Laboratory Incorporated, 207 East Ohio Street, Chicago, Illinois 60611.)

INTERNATIONAL ELECTROTECHNICAL COMMISION (IEC)

Publication 112, 2nd Edition	Recommended Method for Determining the Comparative Tracking Index (CTI) of Solid Insulating Materials Under Moist Conditions

(Application for copies should be addressed to United States of America Standards Institute, Sales Department, 10 East 40th Street, New York, New York 10016.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Qualification. Thermoplastic molding compounds furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.3 and 6.3).

3.1.1 Manufacturers shall obtain qualification approval for each compound. Materials qualified for one color will be qualified for all colors requested by manufacturers.

3.2 Compound. The compound shall consist of all new materials. Requirements for molded parts shall be in accordance with the appendix to MIL-M-14.

3.2.1 Uniformity. All molding compound of the same brand from one manufacturer shall be uniform in texture, in color (see 3.2.2), and in the specified properties as determined by the batch-acceptance inspection specified in 4.4.

3.2.2 Color. Unless otherwise specified in the contract or order, materials shall be furnished in natural color. Other types of compounds shall be supplied in the color specified by the procuring activity (see 6.2). In such cases, if the color is not specified, the compound shall be furnished in the normal color in which manufactured.1/ Natural-colored compounds are those in which no coloring matter of any sort has been added. In evaluating uniformity of color, consideration shall be given to the fact that variation in color of raw materials, may be reflected in the color of the "natural" molding compound. Coloring matters reducing the electrical properties below the specified limits shall not be used.

1/ The natural color of the various types of molding compounds approximate the following color chips of FED-STD-595:

Types	MIL-STD-595	Chip number
PT-F	White	37886
GPT-15F	White	37778
GPT-20F	White	37778
GPT-30F	White	37778
GAT-22F	Dark Straw	20318

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Types	MIL-STD-595	Chip number
GET-30F	Light Buff	13522
GMT-40F	Beige	30318
GLT-10F	Brown	20100
GLT-20F	Brown	20122
GLT-30F	Brown	20142
GST-40F	Brown	~~10032~~
GST-40F	Black	16432

3.2.3 Property values. Standard specimens of the compounds shall conform to the property values shown in TABLE II. 17038

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsbility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

å. Qualification tests (see 4.3)

b. Batch-acceptance inspection (see 4.4)

4.3 Qualification tests. Qualification tests shall be conducted at a laboratory satisfactory to the Naval Electronic Systems Command. Qualification tests shall consist of the tests specified in TABLES I and II.

4.3.1 Specimens. Tests shall be conducted on standard test specimens molded by the manufacturer of each compound for which qualification is required. The number of test specimens to be tested shall be as specified in TABLES I and III, and summarized in TABLE IV.

4.3.2 Preparation. Specimens shall be prepared for tests as specified in 4.5.1.

4.4 Batch-acceptance inspection. Batch-acceptance shall be as shown in TABLE III. They shall be conducted at the manufacturer's plant on standard specimens molded by the manufacturer from each batch of compound to be supplied to molders for production of molded parts under this specification. The molding of specimens and the conduct of tests shall be under the supervision of the inspector.

4.4.1 Batch. A batch is a homogeneous unit of finished molding compound manufactured at one time.

4.4.2 Preparation of specimens. Specimens shall be prepared as specified in 4.5.1.

4.4.3 Rejection. Failure to comply with TABLES II and III shall result in rejection of the batch of compound.

4.5 Test procedures.

4.5.1 Standard test specimens.

4.5.1.1 Number. The minimum number of standard test specimens to be tested shall be as specified in TABLES I and III, and summarized in TABLE IV.

4.5.1.2 Form. The form of the standard test specimens shall be as specified in the applicable ASTM standard, and as shown in TABLES I and III, and summarized in TABLE IV.

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TABLE I. Sampling and conditioning for qualification tests.

Property to be tested	Test method		Specimens		Conditioning procedure (see 4.5.1.5)	Unit of value
	ASTM	Modified by	Form and dimensions	Number tested		
Arc resistance	D495	----	4-inch disk, 1/8 inch thick	3	A	Seconds (minimum average)
Compressive strength, 10 percent deflection	D695	----	1 by 1/2 by 1/2 inch	5	E-48/50+C-96/23/50	lb/in^2 (minimum average)
Dielectric constant: At 100 Hz At 10^6 Hz	D150	----	4-inch disk, 1/8 inch thick	3 3 3 3	E-48/50+des E-48/50+D-24/23 E-48/50+des E-48/50+D-24/23	Maximum average
Dissipation factor: At 100 Hz At 10^6 Hz	D150	----	4-inch disk, 1/8 inch thick	3 3 3 3	E-48/50+des E-48/50+D-24/23 E-48/50+des E-48/50+D-24/23	Maximum average
Dielectric strength: Short-time test Short-time test	D149	4.5.2.1 & 4.5.2.1.1	4-inch disk, 1/8 inch thick 4-inch disk, 1/8 inch thick	3 3	E-48/50+C-96/23/50 E-48/50+D-48/50	Volt per mil (minimum average)
Dielectric breakdown: Short-time test Short-time test	D149	4.5.2.1.2	4-inch disk, 1/8 inch thick	3 3	E-48/50+C-96/23/50 E-48/50+D-48/50	Kilovolt (minimum average)
Flame resistance: Ignition time Burning time	D229	4.6	5-inch bar, 1/2 by 1/2 inch	5	A	Seconds (minimum average) Seconds (maximum average)
Flexural strength	D790	4.5.2.2	5-inch bar, 1/4 by 1/2 inch	5	E-48/50+C-96/23/50	lb/in^2 (minimum average)
Flammability	--	4.7	5-inch bar, 1/2 x thickness	5	(See 4.7)	(See 4.7)
Heat-deflection temperature	D648	4.5.2.3	5-inch bar, 1/2 by 1/4 inch	3	A	Degrees F (minimum average)
Impact strength	D256	----	As per ASTM D256	5	E-48/50+C-96/23/50	ft-lb/in notch (minimum average)

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TABLE I. Sampling and conditioning for qualification tests. - Continued.

Property to be tested	Test method		Specimens		Conditioning procedure (see 4.5.1.5)	Unit of value
	ASTM	Modified by	Form and dimensions	Number tested		
Tensile strength	D638	----	As per ASTM D638	5	E-48/50+C-96/23/50	lb/in^2 (minimum average)
Water absorbtion	D570	4.5.2.4	2-inch disk, 1/8 inch thick	3	E-24/100+des+D-48/50	Percent (maximum average)
Heat resistance	D790	4.5.2.5	5-inch bar, 1/4 by 1/2 inch	5	E-48/50+C-96/23/50 +E-1/200	Percent flexural strength retained (minimum)
Dimensional stability (high temperature)	----	4.5.2.6	5-inch bar, 1/2 by 1/2 inch	5	(See 4.5.2.6)	Percent (maximum average)
Volume resistance	----	4.5.2.7	4-inch disk, 1/8 inch thick	5	(See 4.5.2.7)	Ohm (minimum individual)
Surface resistance	----	4.5.2.7	4-inch disk, 1/8 inch thick	5	(See 4.5.2.7)	Ohm (minimum individual)
Tracking index	----	4.8	4-inch disk, 1/8 inch thick	3	(See 4.8)	Volt
Toxicity when heated: Carbon dioxide Carbon monoxide Ammonia Aldehydes as H.CHO Cyanide as HCN Oxides of nitrogen as No_2 Hydrogen chloride	----	4.5.2.8	5-inch bar, 1/2 by 1/2 inch	4	A	p/m (maximum average)

Inches	MM
1/8" (.125)	3.18
1/4" (.250)	6.35
1/2" (.500)	12.70
2" (2.00)	50.80
4" (4.00)	101.60
5" (5.00)	127.00

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TABLE II. Property values for qualification tests of thermoplastic molding compounds.

Property to be tested	Value required for each type of compound											Units
	Type PT-F	Type GPT-15F	Type GPT-20F	Type GPT-30F	Type GAT-22F	Type GET-30F	Type GMT-40F	Type GLT-10F	Type GLT-20F	Type GLT-30F	Type GST-40F	
Arc resistance	50	65	60	70	80	70	70	85	85	85	30	Sec
Compressive strength, 10 percent deflection	12,000	13,500	14,000	16,200	13,500	17,500	16,200	19,000	20,000	22,000	20,000	lb/in^2
Dielectric constant:												
At 100 Hz	3.6	3.9	3.9	4.1	3.8	4.1	4.1	3.5	3.5	3.5	4.3	----
At 10^6 Hz	3.4	3.7	3.7	4.0	3.8	4.0	4.0	3.5	3.5	3.5	4.2	----
Dissipation factor:												
100 Hz	0.005	0.005	0.005	0.010	0.002	0.005	0.005	0.002	0.002	0.002	0.002	----
10^6 Hz	0.03	0.03	0.03	0.03	0.02	0.03	0.03	0.007	0.007	0.007	0.002	----
Dielectric strength:												
Short-time test	350	370	375	400	400	400	400	450	450	450	400	Volt/mil
Short-time test	325	340	350	375	375	375	375	430	430	430	375	
Dielectric breakdown:												
Short-time test	43	46	46	50	50	50	50	56	56	56	50	Kilovolt
Short-time test	40	42	42	46	46	46	46	54	54	54	46	
Flame resistance:												
Ignition time	50	55	50	55	55	50	55	50	50	50	45	Sec
Burning time	35	50	50	20	50	20	20	20	20	20	10	
Flexural strength	13,000	19,000	20,000	24,000	20,000	24,000	24,000	26,600	28,500	31,400	22,000	lb/in^2
Flammability	V-0	V-0	V-0	V-0	V-0	V-0	V-0	V-0	V-0	V-0	V-0	----
Heat-deflection temperature	160	360	375	385	300	400	385	400	405	405	500	°F
Impact strength	0.6	0.7	0.8	1.1	1.0	1.1	1.1	1.0	1.4	1.8	1.1	ft-lb/in

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TABLE II. Property values for qualification tests of thermoplastic molding compounds - Continued

Property to be tested	Value required for each type of compound											Units
	Type PT-F	Type GPT-15F	Type GPT-20F	Type GPT-30F	Type GAT-22F	Type GET-30F	Type GMT-40F	Type GLT-10F	Type GLT-20F	Type GLT-30F	Type GST-40F	
Tensile strength	8,000	11,500	12,000	15,000	12,000	15,000	15,000	16,000	18,700	23,000	17,500	lb/in^2
Water absorption	0.40	0.35	0.30	0.30	0.40	0.30	0.30	0.69	0.60	0.55	0.05	Percent
Heat resistance	50	50	50	50	70	50	50	85	85	85	95	Percent
Dimensional stability (high temperature)	0.2	0.2	0.2	0.2	.12	0.2	0.2	0.1	0.1	0.1	0.1	Percent
Volume resistance	$1.0x10^9$	$1.0x10^9$	$1.0x10^9$	$1.0x10^9$	$5x10^{12}$	$1.0x10^9$	$1.0x10^9$	$8x10^{12}$	$6x10^{12}$	$4x10^{12}$	$1.0x10^{12}$	Ohm
Surface resistance	$1.0x10^9$	$1.0x10^9$	$1.0x10^9$	$1.0x10^9$	$5x10^{11}$	$1.0x10^9$	$1.0x10^9$	$5x10^{11}$	$5x10^{11}$	$5x10^{11}$	$1.0x10^{10}$	Ohm
Tracking index	180	180	180	180	125	180	180	135	135	135	180	Volt
Toxicity when heated:												
Carbon dioxide	15,000	15,000	15,000	15,000	10,000	15,000	15,000	5,000	5,000	5,000	1,000	
Carbon monoxide	1,000	1,000	1,000	1,000	500	1,000	1,000	250	250	250	200	
Ammonia	2,500	2,500	2,500	2,500	50	2,500	2,500	20	20	20	100	
Aldehydes as H.CHO	50	50	50	50	50	50	50	20	20	20	20	
Cyanide as HCN	60	60	60	60	50	60	60	20	20	20	0	p/m
Oxides of nitrogen as No_2	100	100	100	100	60	100	100	50	50	50	25	
Hydrogen chloride	100	100	100	100	50	100	100	50	50	50	0	

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TABLE III. Property values for batch acceptance inspection of thermoplastic molding compounds.

Property to be tested	Test method		Specimens		Conditioning procedure (see 4.5.1.5)	Unit of value
	ASTM	Modified by	Form and dimensions	Number tested		
Arc resistance	D495	----	4-inch disk, 1/8 inch thick	3	A	Seconds (minimum average)
Dielectric constant at 1 megahertz	D150	----	4-inch disk, 1/8 inch thick	3	E-48/50+D-24/23	Maximum average
Dissipation factor at 1 megahertz	D150	----	4-inch disk, 1/8 inch thick	3	E-48/50+D-24/23	Maximum average
Dielectric strength short time	D149	4.5.2.1	4-inch disk, 1/8 inch thick	3	E-48/50+D-48/50	Volts per mil (minimum average)
Flexural strength	D790	4.5.2.2	5-inch bar, 1/4 by 1/2 inch	5	E-48/50+C-96/23/50	lb/in^2 (minimum average)
Impact strength	D256	----	As per ASTM D256	5	E-48/50+C-96/23/50	ft-lb/in notch (minimum average)
Water absorption	D570	4.5.2.4	2-inch disk, 1/8 inch thick	3	E-24/100+des+D-48/50	Percent (maximum average)

Property to be tested	Value required for each type of compound											Units
	Type PT-F	Type GPT-15F	Type GPT-20F	Type GPT-30F	Type GAT-22F	Type GET-30F	Type GMT-40F	Type GLT-10F	Type GLT-20F	Type GLT-30F	Type GST-40F	
Arc resistance	50	65	70	70	80	70	70	85	85	85	30	Sec
Dielectric constant at 1 megahertz	3.4	3.7	3.7	4.0	3.8	4.0	4.0	3.5	3.5	3.5	4.0	
Dissipation factor at 1 megahertz	0.03	0.03	0.03	0.03	0.02	0.03	0.03	0.007	0.007	0.007	0.002	
Dielectric strength short time	350	370	375	400	400	400	400	450	450	450	400	Volts/mil
Flexural strength	13,000	19,000	20,000	24,000	20,000	24,000	24,000	26,600	28,500	31,400	22,000	lb/in^2
Impact strength, side	0.60	0.70	0.8	1.10	1.0	1.1	1.1	1.0	1.4	1.8	1.1	ft-lb/in
Water absorption	0.40	0.35	0.30	0.30	0.40	0.30	0.30	0.69	0.60	0.55	0.05	Percent

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TABLE IV. Standard test specimens for qualification tests.

Specimen designation	Specimen description	Number of specimens for each type	1/ Inches	MM
1	4 inch disk by 1/8 inch	52	1/8" (.125)	3.18
2	1 inch by 1/2 inch by 1/2 inch	5	1/4" (.250)	6.35
3	2 inch disk by 1/8 inch	3	1/2" (.500)	12.70
4	5 inch by 1/2 inch by 1/2 inch	14	1" (1.00)	25.40
5	5 inch by 1/2 inch by 1/4 inch	13	2" (2.00)	50.80
6	2-1/2 inch by 1/2 inch by 1/2 inch (ASTM D256)	5	2-1/2" (2.50)	63.50
7	ASTM D638	5	4" (4.00)	101.60
8	5 inch by 1/2 inch by thickness	5	5" (5.00)	127.00

4.5.1.3 Molding of test specimens. Test specimens shall be molded by methods which are representative of commercial practice. No special treatment shall be used to improve the properties of the specimens when compared with parts molded in commercial production.

4.5.1.4 Tolerance. Tolerance on dimensions shall be ±5.0 percent unless otherwise specified (see 6.2).

4.5.1.5 Conditioning. Standard test specimens shall be conditioned before test, as specified in TABLES I and III.

4.5.1.5.1 Nomenclature. The following letters shall be used to indicate the respective general conditioning procedures:

Condition A - As received; no special conditioning.
Condition C - Humidity conditioning.
Condition D - Immersion conditioning is distilled water.
Condition E - Temperature conditioning.
Condition desiccation - Desiccation condition, cooling over silica gel or calcium chloride in a desiccator at 23° Celsius (°C) for 16 to 20 hours after temperature conditioning.

4.5.1.5.2 Designation. Conditioning procedures shall be designated as follows:

a. A capital letter indicating the general condition of the specimen; that is, as received, humidity, immersion, or temperature conditioning

b. A number indicating in hours the duration of the conditioning.

c. A number indicating in °C the conditioning temperature.

d. A number indicating relative humidity, whenever relative humidity is controlled.

The numbers shall be separate from each other by slant marks, and from the capital letter by a dash. A sequence of conditions shall be denoted by use of a plus (+) sign between successive conditions.

Examples:
Condition C-96/23/50 - Humidity condition, 96 hours at 23° ±1.1°C and 50 ±2 percent relative humidity.
Condition D-48/50 - Immersion condition, 48 hours at 50° +1°C.
Condition E-48/50 - Temperature condition, 48 hours at 50° +3°C.
Condition E-48/50+C-96/23/50 - Temperature condition, 48 hours at 50° ±3°C followed by humidity condition, 96 hours at 23° +1.1°C and 50 ±2 percent relative humidity.

1/ Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.

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4.5.2 Methods of test. All test measurements with the exception of dielectric strength, dielectric breakdown, heat resistance, and volume and surface resistance (see 4.5.2.1, 4.5.2.1.3, 4.5.2.5, and 4.5.2.7) shall be taken at standard laboratory atmosphere of 23° ±1.1°C and 50 ±2 percent relative humidity. The test methods shall be in conformance with the applicable ASTM (see TABLES I and III) modified as follows:

4.5.2.1 Dielectric strength. The apparatus and procedure specified in ASTM D149 shall be used. Tests shall be made under oil at a frequency not exceeding 100 hertz (Hz), at the temperature of the final conditioning.

4.5.2.1.1 Short-time test. The voltage shall be increased uniformly at the rate of 500 volts per second.

4.5.2.1.2 Dielectric breakdown. The apparatus and procedure specified in ASTM D149 shall be used. The electrodes shall be brass or stainless steel cylinders one inch in diameter by one inch long with the edges rounded to a 1/8-inch radius. Tests shall be made under oil at a frequency not exceeding 100 Hz at the temperature of the final conditioning.

4.5.2.2 Flexural strength. ASTM D790, method 1, procedure A shall be used. The span-depth ratio shall be 16 to 1.

4.5.2.3 Heat-deflection temperature. ASTM D648 shall be used. The specimens shall be tested directly in an oil bath and not in air. The stress load shall be 264 lb/in^2.

4.5.2.4 Water absorption. ASTM D570 shall be used, modified as follows:

a. The specimens shall be conditioned at 100° ±2°C for 24 hours, followed by 16 to 20 hour period of cooling over silica gel or calcium chloride in a desiccator at 23° ±1.1°C.

b. The specimens shall be immersed in distilled water maintained at a temperature of 50° ±1°C for 48 hours. The report shall include only the following data: The percentage increase in weight during immersion calculated to the nearest 0.01 percent as follows:

$$\text{Increase in weight (percent)} = \frac{\text{wet weight-conditioned weight}}{\text{conditioned weight}} \times 100$$

4.5.2.5 Heat resistance. The method shall be the same as specified in 4.5.2.2 except that:

a. Specimens of type PT-F shall be given an additional conditioning of 1 hour at 150° ±2°C and then be tested at 23° ±2°C.

b. Specimens of types GPT-15F, GPT-20F, GPT-30F, GAT-22F, GET-30F, GMT-40F, GLT-10F, GLT-20F, GLT-30F and GST-40F shall be given an additional conditioning of 1 hour at 200° ± 2°C and then be tested at 23° ±2°C.

The average of five such determinations divided by the average flexural strength determined as in 4.5.2.2 shall be multiplied by 100 and recorded as percent flexural strength retained.

4.5.2.6 Dimensional stability (high temperature). The specimens shall be machined so the 1/2 by 1/2 inch ends are smooth and parallel. The specimens shall be subjected to the condition C-96/23/50 (see 4.5.1.5.2). The initial length of the specimens shall then be measured to the nearest 0.001 inch. The specimens shall then be subjected to 10 cycles, each cycle as follows: 48 hours in a circulating air oven at 125° ±5°C plus 24 hours at 23° ±1.1°C and 50 ±2 percent relative humidity. At the completion of 10 cycles the final length of the specimens shall be measured to the nearest 0.001 inch. The percentage dimensional change is calculated to the nearest 0.1 percent as follows:

$$\text{Dimensional change (percent)} = \frac{\text{(initial length - final length)}}{\text{initial length}} \times 100$$

The average percent dimensional change of the five specimens shall be recorded.

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4.5.2.7 Volume and surface resistance - condition C - 720/70/100+dew.

4.5.2.7.1 Specimens. Five 4-inch diameter 1/8-inch thick specimens shall be used. Specimens shall be cleaned by noninjurious methods to assure freedom from contamination. Precautions shall be taken in handling the specimens to avoid additional contamination. Five specimens shall be employed for the measurements of this test.

4.5.2.7.2 Electrodes. Electrodes shall consist of a guarded electrode 2 inches in diameter, 1/4-inch guard ring spaced 1/4 inch from the guarded electrode on the same side and the third electrode 3 inches in diameter on the opposite side and concentric with the guarded electrode. Dimensions of electrodes shall be maintained to a tolerance of plus or minus 1/64 inch. Silver paint, permeable to moisture (Dupont No. 4517 or equal) shall be used for painting electrodes on the specimens. The electrodes shall exhibit a resistance of not more than 5 ohms both before and after the C-720/70/100+dew conditioning when measured with potentials of not greater than 3 volts between points diametrically opposite on each electrode. The specimens shall be permitted to air dry after painting for at least one week in an atmosphere of less than 60 percent relative humidity at a temperature of 25° ±5°C.

4.5.2.7.3 Humidity chamber. The humidity chamber shall consist of a glass container with a corrosion resistant cover. The cover shall be provided with through-panel type insulators. The insulators may serve as supports for the electrode holders as shown in FIGURE 1. The chamber shall be of such size that the ratio of specimens surface area to water surface area shall not exceed 2.5. The ratio of volume of air in the humidity chamber to surface area of the water shall not exceed 10. One hundred percent relative humidity with condensation shall be obtained by natural evaporation from a quantity of distilled water located at the bottom of the chamber. The cover shall be sealed to the chamber by an inert sealing compound applied to the exterior points formed by the cover and the walls of the chamber. A small vent hole shall be provided in the cover to equalize pressure. The vent hole shall be sealed as soon as the air temperature in the humidity chamber has reached 70°C.

4.5.2.7.4 Specimen holders. The specimens shall be installed in a vertical plane in the conditioning chamber with the lower edge of the specimen not closer than 1 inch from the surface of the water. The specimens may be held in position by the electrode contactors in a manner similar to that shown on FIGURE 1. Electrical connection to the specimen holders may be made by means of the through-panel insulators. The insulators shall be capable of withstanding the adverse conditions within the chamber without excessive loss of insulating properties. (Insulator resistance to cover plate shall at all times exceed 10 megohms (M Ω).) Polytetrafluoroethylene insulators on the humidity side of the conditioning chamber are recommended to meet this requirement. These should be cleaned with alcohol before the start of each test. Electrode contactors and all other metallic parts of the sample shall be silver plated. Contact pressure against the electrodes may be provided by backing the contactors with phosphor bronze springs or other corrosion-resistant spring material.

4.5.2.7.5 Heating chamber. The humidity chamber shall be installed in an oven or other heating chamber capable of maintaining a temperature of 70° ±1°C. The rate of heating of the oven shall be such that the air temperature at a point near the volumetric center of the humidity chamber shall attain 70°C in 4 hours ±1 hour. The quantity of water in the chamber shall be such that the water temperature shall attain 65°C in 4 hours ±1 hour. Room temperature shall be maintained at 25° ±5°C. The insulation of the conductors connecting the through-panel insulators to the measuring equipment shall not be significantly deteriorated by the elevated temperatures encountered in the oven. Polytetrafluoroethylene coated wire is recommended.

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4.5.2.7.6 Measurements.2/ Volume and surface resistances shall be measured by the three terminal method employing measuring equipment such as a megohm bridge capable of applying 500 volts direct current (d.c.) to the specimen. A single set of measurements shall be made of each specimen while in the conditioning chamber after 30 days of the specified conditioning. Conversion of the measurements to resistivities is not required since electrode dimensions are specified. The potentials shall be applied to the specimens as shown on FIGURE 2 or with polarities opposite to those shown on the figure. Surface resistance measurements shall be made on the same specimens as those used for volume resistance except that the potentials of guard and low electrodes shall be interchanged. The volume and surface resistance shall be measured in each case, 1 minute after the potentials are applied. Low values of volume and surface resistance (below 5 M Ω) may be measured by the circuits shown on FIGURE 3.

4.5.2.8 Toxicity when heated. The method described in AD297457 shall be used.

4.6 Flame resistance. Flame resistance shall be determined in accordance with method II of ASTM D229 with the following exceptions:

a. Determination of weight and weight loss, such as in definitions, apparatus, procedures, calculations, and report sections, are not applicable.

b. Flame cabinet. The 9/16-inch slot at bottom of the flame cabinet shall be on all four sides. The door shall be provided with a 1-1/4 inch diameter peep hole located directly opposite the heater coil when the door is closed. The hole shall be kept closed during testing by means of a cover.

c. Pyrometer. The means of correction from black-body radiation to actual conditions of this test shall be delineated as follows:

1. When a pyrometer calibrated for black-body emission is used, 6°C shall be added to the pyrometer to obtain the true temperature of the Nichrome V coil.

d. Specimens. Test specimens shall be as follows:

1. Specimens shall be molded to 1/2 by 1/2 by 5 ±1/16 inches.

2. The test sample shall consist of five test specimens.

e. Calibration. In the calibration of this equipment, adjust the heater current to obtain an equilibrium temperature of 860° ±2°C.

f. Calculation of burning time. Arrange the five values of burning time in increasing order of magnitude, as T_1, T_2, T_3, T_4, and T_5. Compute the following ratios:

$$\frac{T_2 - T_1}{T_5 - T_1} \quad \text{and} \quad \frac{T_5 - T_4}{T_5 - T_1}$$

If either of these ratios exceeds 0.642, then T_1 or T_5 is judged to be abnormal and is eliminated. The burning time reported shall be the average of the remaining four values.

g. Average ignition time. The ignition time is calculated as the arithmetic mean of the five specimens.

2/ Because of the variability of the resistance of a given specimen with test conditions and because of non-uniformity of the same material from specimen to specimen, determinations are usually not reproducible to closer than 10 percent and are often even more widely divergent. A range of values of 10 to 1 may be obtained under apparently identical conditions. Errors in resistance determinations may result from the fact that the current measuring device is shunted by the resistance between the guarded terminal and the guard system. In some bridge techniques a standard resistor in the bridge is shunted by the resistance between the unguarded terminal and the guard system. To assure validity of the volume and surface resistance measurements obtained by the bridge methods the resistance between the unguarded and the guard terminal should be at least five times greater than the standard resistance employed in the bridge. This may be ascertained by direct two terminal measurements between these two terminals.

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4.7 Flammability. Flammability rating shall be determined in accordance with Underwriter Laboratories Standard 94. Result shall be reported as V-0.

4.8 Comparative tracking index (CTI). The CTI shall be determined in accordance with the International Electrotechnical Commission (IEC), Publication 112, 2nd Edition - Method for Determining the Comparative Tracking Index of Solid Insulation Materials Under Moist Conditions.

5. PACKAGING

5.1 Not applicable to this specification. (For finished molded parts see section 50 of the appendix to MIL-M-14 for the applicable preparation for delivery requirements.)

6. NOTES

6.1 Intended use. The thermoplastic compounds covered by this specification are general purpose types intended for applications requiring good electrical and mechanical properties. The compounds are used in electronic type connectors and similar applications.

6.1.1 Flame resistant types of thermoplastic compounds. Flame retardants reduce arc tracking resistance of the compounds. Do not use in applications above 250 volts.

6.2 Ordering data. Procurement documents should specify:

6.2.1 Acquisition requirements.

a. Title, number, and date of this specification.

b. Type required (see 1.2).

c. Color required (see 3.2.2).

d. Tolerance other than ±5.0 percent (see 4.5.1.4).

6.3 With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List QPL-24519 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is Naval Electronic Systems Command, ELEX 81112, Department of the Navy, Washington, DC 20363, and information pertaining to qualification of products may be obtained from that activity. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (see 6.3.1).

6.3.1 Copies of "Provisions Governing Qualification SD-6" may be obtained upon application to Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

User activities:
Navy - SH, AS, MC

Preparing activity:
Navy - EC

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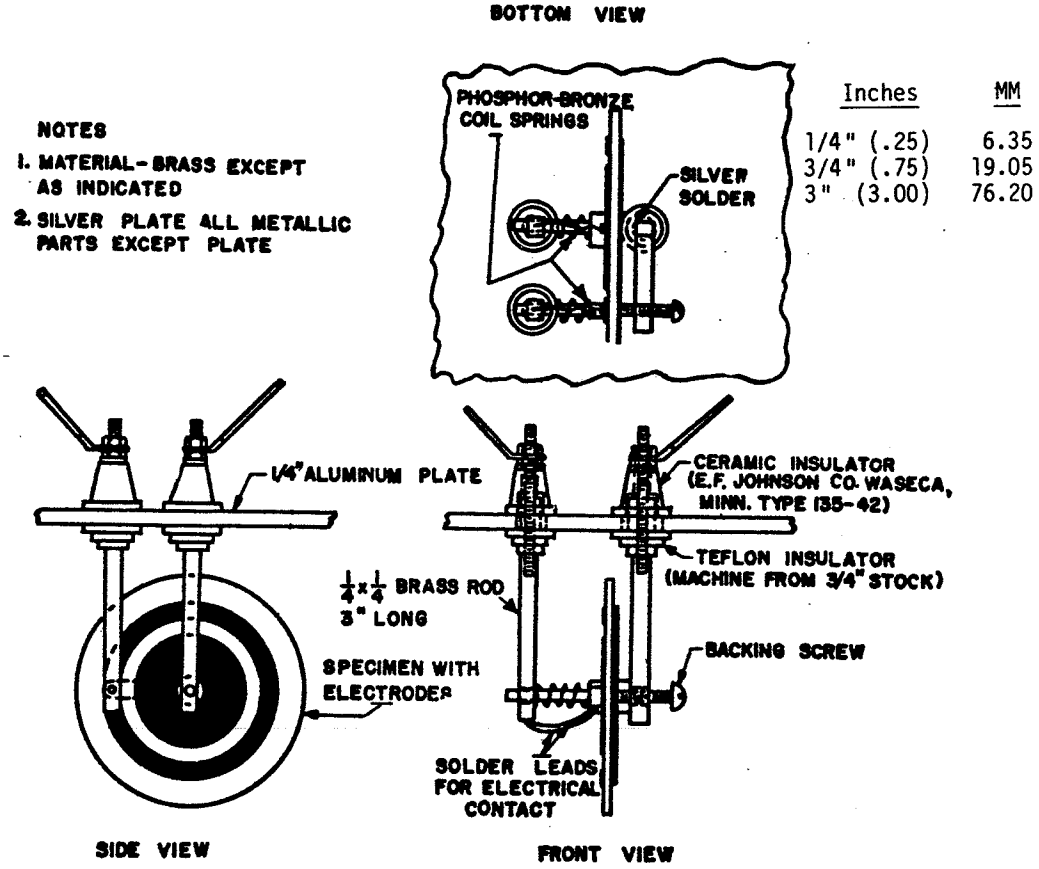


FIGURE 1. Specimen holders, electrodes, test samples, and humidity chamber cover - volume and surface resistance test.

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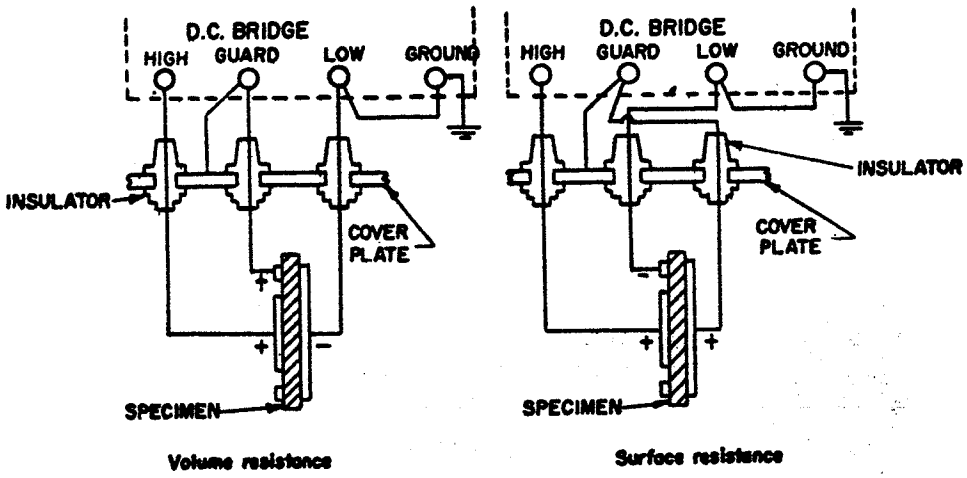


FIGURE 2. Arrangements for volume resistance and surface resistance test.

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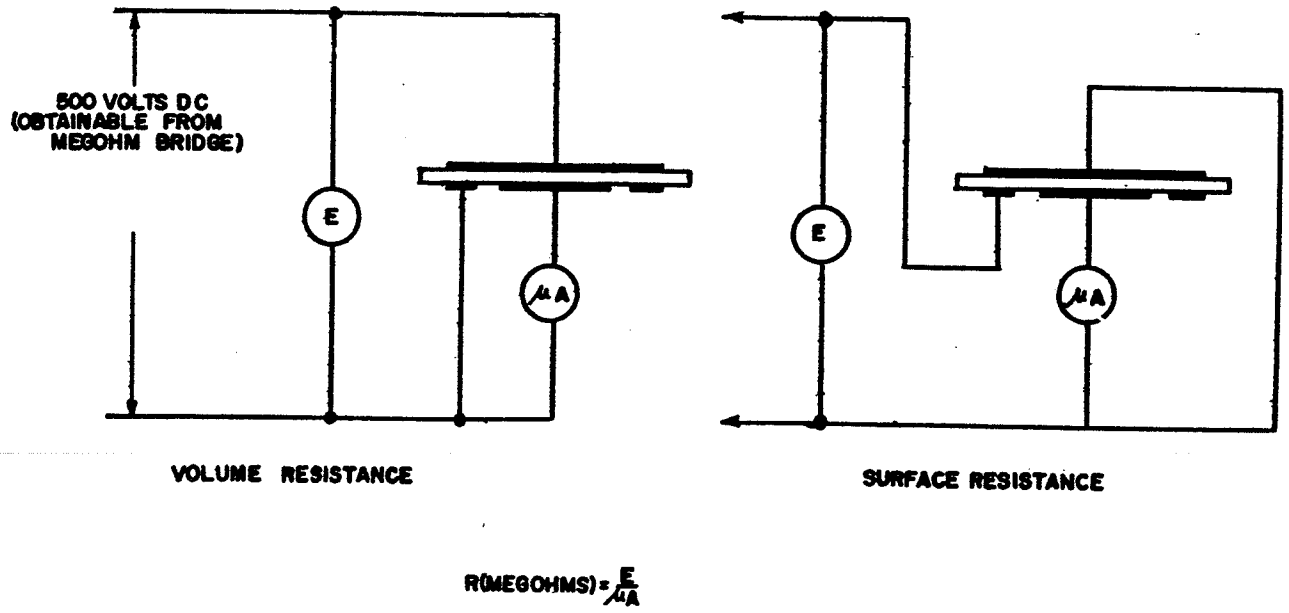


FIGURE 3. Circuits for measuring low values of volume and surface resistance.

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3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
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5. PROBLEM AREAS

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c. Reason/Rationale for Recommendation:

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